Basic Programming
Practice Question Bank

Consolidated coding practice problems for fresher placement preparation

Numbers · Arrays · Strings · Sub-Strings · Dictionary/Map · HackerRank 5☆Practice

Numbers

1. Wap to print all numbers from 1 to 100.

2. Wap to print all even numbers from 0 to 100.

3. Wap to print all the factors of a given number.

4. Wap to print all the multiples of a given number.

5. Write a program to print all the odd numbers in one line and all even numbers in another line.

6. Wap to print the total sum of integers from 1 to a given number.
Example: Input: 10 → Output: 55 (1+2+3+...+10 = 55)

7. Wap to check whether the given number is a perfect number or not, without defining any function.

8. Wap to check whether the given number is a perfect number or not, by defining a function.

9. Wap to print all perfect numbers from 1 to 10000.

10. Wap to reverse the given number without using string methods.

11. Wap to check whether the given number is a palindrome or not.

12. Wap to print all palindrome numbers from 1 to 1000.

13. Wap to count the number of factors of a given number.

14. Wap to check whether the given number is a prime number or not.

15. Wap to print all the prime numbers from 1 to 1000.

16. Wap to check whether the given number is an Armstrong number or not.

17. Wap to print all the Armstrong numbers from 1 to 10000.

18. Wap to check whether the given number is a Disarium number or not.

19. Wap to print all the Disarium numbers from 1 to 10000.

20. Wap to check whether the given number is a Happy number or not.

21. Wap to print all the Happy numbers from 1 to 10000.

22. Wap to get the HCF of given numbers.

23. Wap to get the LCM of given numbers.

24. Wap to print the factorial of a given number.

25. Wap to check whether the given number is a Krishnamurthy number or not.

26. Wap to print the Fibonacci series up to N terms.

27. Wap to check whether a given number belongs to the Fibonacci series.

28. Wap to find the sum of digits of a given number.

29. Wap to find the product of digits of a given number.

30. Wap to count the number of digits in a given number.

31. Wap to find the largest and smallest digit in a given number.

32. Wap to check whether a given number is a Strong number or not (sum of the factorials of its digits equals the number itself, e.g., 145 = 1! + 4! + 5!).

33. Wap to check whether a given number is an Automorphic number or not (its square ends with the number itself, e.g., $25^2 = 625$).

34. Wap to check whether a given number is a Neon number or not (sum of the digits of its square equals the number itself, e.g., $9^2 = 81$, 8+1 = 9).

35. Wap to swap two numbers without using a third (temporary) variable.

36. Wap to convert a decimal number to its binary, octal, and hexadecimal equivalents.

37. Wap to convert a binary number to its decimal equivalent.

Multiple-Condition Number Problems

1. Next prime with prime digit sum.

Given: N. Find the smallest prime number greater than N such that:

- The number itself is a prime.
- The sum of its digits is also a prime number.

Example: Input 10 → Output 11 | Input 22 → Output 23

2. Given: N. Find the smallest number greater than N such that:

- The number is divisible by 7.
- The sum of its digits is a prime number.
- The number contains '3' at least once.

Example: Input 12 → Output 133 | Input 133 → Output 203

3. Given: N. Find the smallest number greater than N such that:

- The number is a palindrome.
- The sum of its digits is an even number.

Example: Input 20 → Output 22 | Input 131 → Output 141

4. Given: N. Find the smallest number greater than N such that:

- The number is divisible by 5.
- The number ends with digit '0'.
- The sum of its digits is an even number.

Example: Input 13 → Output 20 | Input 94 → Output 110

5. Given: N. Find the smallest number greater than N such that:

- The number is a palindrome.
- The sum of its digits is divisible by 3.
- The number is odd.

Example: Input 20 → Output 33 | Input 130 → Output 141

6. Given: N. Find the smallest number greater than N such that:

- The number is divisible by 9.
- The number contains digit '5' at least once.
- The product of its digits is divisible by 5.

Example: Input 40 → Output 45 | Input 131 → Output 153

7. Given: N. Find the smallest number greater than N such that:

- The number is divisible by 11.
- The sum of its digits is an odd number.
- The number ends with digit '1'.

Example: Input 20 → Output 121 | Input 131 → Output 341

8. Given: N. Find the smallest number greater than N such that:

- The number is a palindrome.

- The number is divisible by 3.
- The sum of its digits is an even number.

Example: Input 20 → Output 66 | Input 131 → Output 141

9. Given: N. Find the smallest number greater than N such that:

- The number is an Armstrong number.
- The number is even.
- The sum of its digits is also an even number.

Example: Input 258 → Output 370 | Input 105 → Output 370

10. Given: N. Find the smallest number greater than N such that:

- The number is divisible by both 3 and 7.
- The number ends with digit '1'.
- The sum of its digits is divisible by 3.

Example: Input 20 → Output 21 | Input 100 → Output 231

11. Given: N. Find the largest number smaller than N such that:

- The number is a prime.
- The sum of its digits is a prime number.
- The number contains digit '7' at least once.

Example: Input 100 → Output 67 | Input 200 → Output 197

12. Given: N. Find the largest number smaller than N such that:

- The number is divisible by 9.
- The number is a palindrome number.
- The sum of its digits is an even number.

Example: Input 200 → Output 99 | Input 500 → Output 99

13. Given: N. Find the smallest number greater than N such that:

- The number is a perfect square.
- The sum of its digits is an odd number.

Example: Input 10 → Output 16 | Input 17 → Output 25

14. Given: N. Find the largest number smaller than N such that:

- The number is a palindrome.
- The number is divisible by 4.

Example: Input 50 → Output 44 | Input 100 → Output 88

Arrays

In Python, solve these using a list; in Java, use an array or ArrayList as appropriate.

1. Wap to find the largest element in an array.

2. Wap to find the smallest element in an array.

3. Wap to find the 2nd largest element in an array.

4. Wap to find the 2nd smallest element in an array.

5. Wap to reverse an array without using in-built functions.

6. Wap to print the sum of all elements of a given array.

7. Wap to get the average of all array elements.

8. Wap to find duplicate elements in a given array.

9. Wap to remove duplicates from a given array.

10. Wap to print the average of all even numbers in an array.

11. Wap to check if a key is present in a list or not — if present, return its index, else return -1. (Linear Search Algorithm)

12. Wap to rotate an array in the clockwise direction N times.

13. Wap to rotate an array in the anti-clockwise direction N times.

14. Wap for the Binary Search Algorithm.

15. Wap for the Bubble Sort Algorithm.

16. Wap to merge 2 arrays.

17. Wap to find common elements in two arrays.

18. Wap to print the frequency of elements in an array.

19. Wap to print all the sub-arrays of a given array.

20. Wap to find the maximum sum of a sub-array of a given array.

21. Wap to print the count of even and odd numbers in an array.

22. Wap to check if an array is sorted or not, without using in-built functions.

23. Wap to find a pair with a given sum in an array.

24. Wap to find the maximum difference between elements of an array.

25. Wap to find the intersection of two arrays.

26. Wap to find the length of an array.

27. Wap to print only the sub-array with the minimum sum.

28. Wap to find the missing number in a given array.

29. Wap to move all zeroes in an array to the end.

30. Wap for the Merge Sort Algorithm.

31. Wap for the Selection Sort Algorithm.

32. Wap for the Insertion Sort Algorithm.

33. Wap to find the majority element in an array (the element that appears more than n/2 times).

34. Wap to find the Kth largest element in an array.

35. Wap to find the Kth smallest element in an array.

36. Wap to find all 'leaders' in an array (an element is a leader if it is greater than every element to its right).

37. Wap to find the equilibrium index of an array (an index where the sum of elements to its left equals the sum of elements to its right).

38. Wap to segregate positive and negative numbers in an array (rearrange so all negative numbers come before all positive numbers).

39. Wap to find the transpose of a given matrix (2D array).

40. Wap to print a given matrix (2D array) in spiral order.

Strings

In Python, use the str type; in Java, use the String class (StringBuilder where mutation is needed).

1. Wap to reverse a string.

2. Wap to check if a given string is a palindrome or not.

3. Practice all the built-in string functions/methods available in your programming language (e.g., Python's str methods or Java's String class methods).

4. Wap to remove spaces from a given string.

5. Wap to count the characters in a given string.

6. Wap to find duplicate characters in a given string.

7. Wap to remove duplicate characters from a given string.

8. Wap to find the first non-repeated character in a given string.

9. Wap to count the words in a sentence.

10. Wap to find the largest word in a sentence.

11. Wap to reverse each word in a sentence.

12. Wap to find the shortest word in a sentence.

13. Wap to check if a given substring exists in a given string.

14. Wap to count the frequency of characters in a given string.

15. Wap to find the first repeated character in a given string.

16. Wap to convert a whole string to upper case.

17. Wap to convert a string to lower case.

18. Wap to swap the cases of a string.

19. Try the above 3 questions when the string contains spaces.

20. Wap to count vowels and consonants of a string.

21. Wap to sort the characters in a given string.

22. Wap to count the number of vowels, consonants, spaces, numbers, and special characters in a given string.

23. Wap to check if the given string is an Anagram or not, without using any built-in sort function/method.

24. Wap to find the index of a given letter in a string.

25. Wap a program to convert the string 'dhee coding lab' to:
i] Dhee Coding Lab ii] lab coding dhee iii] eehd gnidoc bal
iv] bal gnidoc eehd v] Eehd Gnidoc Bal vi] EehD GnidoC BaL

26. Wap to check if the given string is an Anagram of another, using a built-in sort function/method.

27. Wap to check if one string is a rotation of another string.

28. Wap to compress a string using counts of repeated characters (e.g., 'aabccccaaa' → 'a2b1c5a3').

29. Wap to remove a specific character from a given string.

30. Wap to count the occurrences of a given substring within a string.

31. Wap to check whether a given string contains only digits, only alphabets, or is alphanumeric.

32. Wap to convert a numeric string to an integer, without using any built-in string-to-number conversion function.

33. Wap to check whether the parentheses/brackets in a given string are balanced.

Sub-Strings

1. Wap to print all substrings of a given string.

2. Wap to print all palindromic substrings of a given string.

3. Wap to print the longest palindromic substring.

4. Wap to check whether a given string is a substring of another string, without using the 'in' operator.

5. Wap to find the longest common prefix among a list of strings.

Dictionary / Map (Key-Value Pairs)

In Python use dict; in Java use HashMap / TreeMap / LinkedHashMap as appropriate.

1. Wap to fetch only the even values from a dictionary/map.
Example data — val1→10, val2→20, val3→23, val4→22

2. Wap to build a dictionary/map that stores the ASCII value of each character in a string.
Input: 'abcd' → Output: a→97, b→98, c→99, d→100

3. Wap to build a dictionary/map that stores each character of a string against its index.
Input: 'hello' → Output: 0→h, 1→e, 2→l, 3→l, 4→o

4. Wap to build a dictionary/map from a list containing a mix of strings and numbers, mapping each string to its first and last letter joined together (skip the numeric entries).
Input: hai, 89, 3.4, hello, 90, py → Output: hai→hi, hello→ho, py→py

5. Wap to build a dictionary/map from a sentence, where each word maps to a short code following the pattern shown below.
Input: 'hai hello good morning' → Output: hai→a, hello→l, good→gd, morning→n

6. Wap to group a list of filenames by their extension.
Input: jiocinema.com, file.py, web.html, amazon.com, text.py
Output: com→[jiocinema, amazon], py→[file, text], html→[web]

7. Wap to count the vowels in each word of a sentence and store the result in a dictionary/map.
Input: 'hai hello' → Output: hai→2, hello→2

8. Wap to count the number of occurrences of each character in a string using a dictionary/map and a while loop.

9. Wap to swap the keys and values of a dictionary/map.

10. Wap to merge two dictionaries/maps into one.

11. Wap to find the key with the maximum value in a dictionary/map.

12. Wap to sort a dictionary/map by its values.

13. Wap to count the frequency of each word in a sentence using a dictionary/map.

14. Wap to convert two lists/arrays into a dictionary/map (one as keys, the other as values).

15. Wap to check whether two dictionaries/maps are equal.

16. Wap to remove a key from a dictionary/map only if it exists (without raising an error).

HackerRank Practice(Do these questions to reach 5☆ on Hackerrank)

1. Solve me first
2. Compare the Triplets
3. A very big sum
4. Diagonal Difference
5. Plus Minus
6. Staircase
7. Min-max sum
8. Time conversion
9. Birthday cake candles
10. Grading students
11. Apple and orange
12. Number line jumps
13. Breaking the records
14. Subarray division
15. Divisible sum pairs
16. Migratory birds
17. Bill division
18. Sales by match
19. Drawing book
20. Between two sets
21. Cats and a mouse
22. Day of a programmer
23. Counting valleys
24. The hurdle race
25. Designer PDF viewer
26. Electronics shop
27. Utopian tree
28. Angry professor
29. Beautiful days at the movies
30. Strange advertising
31. Cut the sticks
32. Sequence equation
33. Circular array rotation
34. Find digits
35. Missing numbers
36. Find the median
37. Closest numbers

38. Sherlock and array
39. Jim and the orders
40. Mark and toys
41. Halloween sale
42. Service lane
43. Chocolate feast
44. Fair rations
45. Making anagrams
46. Beautiful binary string
47. Alternate characters
48. Ice cream parlor
49. Pangrams
50. Two Strings
51. Camelcase
52. Caesar Cipher
53. HackerRank in a String!
54. Strong Password